

IVAN BEDNOSTIN

Mechanical Engineering Student | Computer Science Minor

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Education

Shippensburg University — Shippensburg, PA

Bachelor of Science in Mechanical Engineering | Expected May 2028

GPA: 3.5+

Technical Skills

Design & CAD: **SolidWorks**, Blender

Programming & Data: **Python** (pandas, NumPy), **MATLAB**, C, Data Analysis, Machine Learning

Simulation: **Simulink**

Fabrication & Tools: **3D Printing**, Prototyping, Git/GitHub, Excel

Engineering Experience

Shippensburg University — Media Services

Print Operator / 3D Printing Technician | Shippensburg, PA | September 2024 – Present

- **Design, 3D-print, and prototype** custom parts and components from scratch in **SolidWorks**, refining geometry and fit through testing.
- Operate, maintain, and **troubleshoot** 3D printers and office production equipment, including mounting, laminating, and cutting, resolving mechanical and software issues while maintaining **precise alignment** and consistent output quality.

Engineering Research

Undergraduate Researcher — Wind Turbine Yaw-Misalignment Classification

Shippensburg University | Faculty Mentor: Dr. João P. Dias | May 2026 – Present

- Developed a reproducible **Python** workflow to process simulated **OpenFAST/TurbSim** wind-turbine data, including metadata parsing, transient removal, labeling, windowing, statistical feature extraction, and ML-ready dataset preparation.
- Performed exploratory data analysis and **feature engineering** across wind, structural-response, power-production, and rotor-operation signals, using **pandas**, **NumPy**, **Plotly**, and interactive **HTML-based heatmaps**.
- Developed and evaluated binary and multiclass **Logistic Regression** models to classify **yaw-aligned vs. misaligned operation** and distinguish among five yaw-misalignment levels.

Selected Engineering Projects

Plastic-to-Filament Recycling System (Sustainability)

- Converted an Ender-3 Pro V2 into a **portable plastic-to-filament recycling system** by integrating **custom mechanical components**, reconfigured electronics, and **custom control code**.
- Automated filament extrusion and spool winding with controlled temperature and speed for consistent output.

Popsicle-Stick Truss Bridge (Statics)

- Designed and iteratively optimized a pin-jointed truss through **structural analysis** and repeated testing using **online simulation tools**.
- Applied **statics** principles, including **force distribution** and **load-path analysis**, to refine member geometry, achieving an approximately **300:1 strength-to-weight ratio** and **1st place** in the class competition.

Dynamic Suspension System Modeling (Dynamics)

- Derived the governing second-order **differential equation** for a one-degree-of-freedom mass-spring-damper system from Newton's Second Law, then reformulated it as first-order state equations and implemented the numerical solution in **MATLAB** using **ode45**.
- Developed an adjustable **MATLAB** simulation to vary **spring stiffness** and **damping**, comparing displacement, **transient response**, and **resonance** behavior, and extended the analysis with a **Simulink** model of a road pothole response.

Relevant Coursework

Modeling & Simulation (MATLAB), **Dynamics**, Differential Equations, **Programming for Engineers (C)**, Probability & Statistics, **Engineering Graphics (SolidWorks)**